What kind of company do words keep?
Revisiting the distributional semantics of J.R. Firth & Zellig Harris

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Abstract

The power of word embeddings is attributed to the linguistic theory that similar words will appear in similar contexts. This idea is specifically invoked by noting that "words will be known by the company they keep," a quote from British linguist J.R. Firth who, along with his American colleague Zellig Harris, is often credited with the invention of "distributional semantics." While both Firth and Harris are cited in all major NLP textbooks and many foundational papers, the content and differences between their theories is seldom discussed. Engaging in a close reading with their work, we discover two distinct and in many ways divergent theories of meaning. One is "introverted," focused on the internal workings of language, while the other is "extroverted" and invites us to consider words in new company, not just with other linguistic elements but also in a broader cultural context. Contrasting these theories from the perspective of current debates in NLP, we discover in Firth a figure who could guide the field towards a more culturally grounded notion of semantics. We consider how an expanded notion of "context" might be modeled in practice through two different strategies: comparative stratification and syntagmatic extension.

1 Introduction

> We are in the world and the world is in us.
>
> Alfred North Whitehead
> (1968, 165; cited in Firth 1957b, 29)

If you have read any papers in computational linguistics in the past thirty years, it is likely that you have come upon the following quote from British linguist J.R.Firth: "words shall be known by the company they keep." Cited in most major textbooks (Eisenstein, 2019; Jurafsky and Martin, 2021; Russell and Norvig, 2020; Manning and Schütze, 1999), several foundational papers, and hundreds of other NLP articles, these words have come to index a theoretical orientation in a field that is increasingly focused on computation, often at the expense of linguistic theory (on these trends in NLP see Church and Liberman, 2021; Halevy et al., 2009; Henderson, 2020; Manning, 2015; Norvig, 2012). Together with American linguist Zellig Harris, Firth is often called upon to justify a distributional theory of semantics, where the meaning of different lexical units is conceived in terms of relative co-occurrence and shared contexts (Sahlgren, 2008).

While Harris and Firth are often invoked, their ideas are seldom closely engaged. Hailing from very disparate traditions, Harris and Firth had radically different ideas on the scope and context of linguistic analysis and probably would not have agreed even on the question of distributional semantics. Extending on the information theory of Claude Shannon (1948), Harris was determined to work out a structuralist theory of language in terms of mathematical information (Léon, 2011; Nevin, 1993). Firth, on the other hand, came to linguistics through anthropology and borrowed heavily from pragmatic philosophies of language. For him (1957b), linguistic analysis always started with the "context of situation" and necessarily accounted for non-verbal actors and objects.

Considering the definition and extent of linguistic context is important for many reasons, as a spate of recent publications suggests (Bisk et al., 2020; Bender and Koller, 2020; Trott et al., 2020; Tamari et al., 2020; Hovy, 2018). First of all, it touches upon the limits of current paradigms in NLP, where corpus linguistics is perfected through increasingly complex models trained on increasingly massive corpora (Bender et al., 2021). This approach may advance the identification of linguistic *form*, but might ultimately have little to say about the relation of *meaning* to the social world (Bender and Koller, 2020; Bisk et al., 2020). Secondly, even with more modest ambitions, several NLP applications—e.g.

with spatial (McKenzie and Adams, 2021) or historical (Kutuzov et al., 2018) data—require that linguistic patterns are related to other types of structure. Thirdly, from the perspective of sociocultural research, meaning intrinsically varies as language is used in different settings and indexed to different social categories (Labov, 1972; Silverstein, 2003; Eckert, 2018; Bourdieu, 1984; Hovy, 2018). Finally, without a broader sense of context, NLP and language modeling in particular remains trapped in a paradigm where language is always treated as universal, making invisible both different speech communities (e.g. Nguyen et al., 2021) and the biases of language (e.g. Lu et al., 2020; Blodgett et al., 2020; Sap et al., 2020).

After a brief history of distributional semantics, we outline work that Harris and Firth did on distribution and, more broadly, on the scope of linguistic analysis. We look at some of the ways in which NLP has tried to account for broader context within the distributional paradigm. We suggest that existing strategies can be understood in terms of either "comparative stratification" or "syntagmatic extension." We conclude with thoughts on how re-reading Harris and, in particular, Firth might aid the field of NLP with its current aporias. If words shall be known by the company they keep, then the question follows: what company do they keep? Are they found only with words and other linguistic elements, or do they mingle with other types of entities? Or, as Firth himself wrote: "Many different answers could be given to the question 'Distribution of what, where and how?"' (Firth, 1957a, v).

2 Background: Distributional Semantics and NLP

Distributional semantics has been an integral part of computational linguistics since the beginnings of the field, but in a discontinuous manner that encompasses at least two distinct eras. Firstly, during the 1950s and 60s, Harris was integral to the mathematization of linguistics after the Second World War (Rubenstein and Goodenough, 1965; Léon, 2021). Firth was skeptical of efforts to mechanize linguistics, but he nonetheless consulted for some of the early work on machine translation at Cambridge (Léon, 2007, 410). Secondly, when computational linguistics returned to its "empiricist" roots in probabilistic methods and information theory in the mid-80s and early 90s (Norvig 2012, 30; Léon 2021, 141), Firth and Harris were alongside Shannon among the authors who were evoked, in an ACL "Special Issue on Computational Linguistics Using Large Corpora," as foundational figures of a tradition that had been overshadowed for decades by the "rationalism" of characters such as Noam Chomsky and Marvin Minsky (Church and Mercer, 1993, 15).

During this "corpus turn," the rapid automation of linguistics was driven by a resumed connection with postwar computational linguistics and information theory (Léon, 2021, 3). However, important papers on non-neural lexical semantics from the era (Landauer and Dumais, 1997; Deerwester et al., 1989, 1990) and other vector space models or their applications (Schutze, 1992; Schütze, 1993; Schütze and Pedersen, 1993; Burgess, 1998; Blei et al., 2002) generally did not cite Firth or Harris. In short, while Firth and Harris were not regularly used as stand-ins for linguistic theory during this time of "count based" models (Baroni et al., 2014), a general revival of empiricism and distributional approaches to meaning did signal a potential resurgence of interest in their thinking.

At the turn of the millennium, computational approaches to lexical semantics shifted toward predictive neural language modeling (Henderson, 2020). During the 2000s, the application of neural networks to language modeling tasks (e.g. Bengio et al., 2003) and the development of self-supervision techniques (e.g. Raina et al., 2007) set the stage for the word embedding breakthroughs of the early 2010s (e.g. Mikolov et al., 2013). By the end of the decade, the introduction of Long Short Term Memory networks (LSTMs) (e.g. Graves et al., 2013) and then of the Transformer model (Vaswani et al., 2017) had made way for the next breakthrough, the large-language modeling revolution (e.g. Devlin et al., 2019; Peters et al., 2018).

Following the introduction of the word2vec model and its powerful but "static" embeddings, Harris in particular was frequently cited (Bojanowski et al., 2017; Le and Mikolov, 2014; Levy and Goldberg, 2014; Levy et al., 2015), often (but not always) along with Firth (Goldberg, 2017; Jurafsky and Martin, 2009, 2021; Hamilton et al., 2016; Eisenstein, 2019; Lin and Dyer, 2010; Bruni et al., 2014). However, despite an explosion of citations (Bisk et al., 2020, 8719), this interest has not been very engaged. In fact, the canonization of Firth and Harris during this time is paradoxical.

On the one hand it seems that they are invoked to lend theoretical authority to a field that struggles to lift its gaze from the latest state-of-the-art numbers (Manning, 2015; Bender and Koller, 2020). Yet, on the other hand, the unspoken conclusion from the ascent of connectionism and the language modeling revolution in particular was that "learning from data made linguistic theories irrelevant" (Henderson, 2020, 6295). In other words, just as NLP seemed to lose interest with linguistic theory, it elevated two pioneering theoreticians to canonical status, but without deeply engaging with their work. Firth and Harris become figures who justify a relatively narrow conception of meaning, one that is predominantly intra-linguistic, without much to say about its uses in the world.

Though Léon discusses the contrast between Harris and Firth in the context of corpus linguistics (2008) and their influence on the history of computational linguistics (2021), her work does not address the differences in their distributional theories and conceptions of "context," nor the renewed and paradoxical significance of the two authors for language modeling. In our contribution, we emphasize the gap between the ideas of Firth and Harris as well as the potential insights a re-reading of their work offer for expanding the scope of computational semantics.

3 Harris's distributional structuralism

Few linguists contributed more to linguistic theory than Zellig Harris (1909–1992), and not just for serving as Chomsky's doctoral advisor. In fact, the two came to share little in common (Goldsmith, 2005; Nevin, 2010); whereas Chomsky's generative grammar repositioned linguistics as a cognitive science seeking to understand, in so few words, the idealized mental representations and structures enabling language acquisition and production (e.g. Chomsky, 1972), Harris' radically distributional approach to language saw the natural language corpus as the sole starting point from which a general linguistic theory could arise.

This theory consisted of a linguistic structure best understood as segmentable into a finite set of formal objects characterized by structured, constrained patterns of correspondence with one another (Harris, 1951, 1954, 1991). Such patterns of correspondence can be observed only in language-in-use, that is, in natural language corpora. In his foundational paper "Distributional Structure," for example, he provides a purely distributional account of how one might induce the semantic meanings of oculist, eye-doctor, and lawyer from the partial (in the case of oculist and lawyer) or nearly complete (in the case of oculist and eye-doctor) overlap in their observed "environments" of use (Harris, 1954, 156–157). This general approach applies to other levels of linguistic analysis, such as morphophonemics (e.g., Harris, 1954, 155). Rather than producing a series of descriptive rules for the distribution of each phoneme, morpheme, or word, the goal was to achieve greater parsimony by grouping these elements into structurally equivalent classes sharing the same distributional rules, compounding elements in a hierarchical manner.

In this section we draw attention to three aspects of Harris's distributional linguistics: the relationship it posits between meaning and form; assumptions about heterogeneity among speakers of the same dialect; and the concept of sublanguages.

3.1 Meaning and form

A result of this vision of a linguistics—concerned above all with the structural and probabilistic constraints governing the combination of formal elements—is that the discipline would be fully autonomous, not only from biology and psychology but even from semantics, phonetics, and logic, "complete without intrusion of other features such as history or meaning" (Harris 1954, 146; Goldsmith 2005, 725–726). Harris's reasoning drew on the particular status he gave to linguistics among all the sciences. Taking language as its object of inquiry, linguistics lacks—unlike other sciences—a *metalanguage* external to language, i.e., to its object of inquiry (Nevin, 1993, 358–359). Even if some sort of symbolic system is used, "those symbols will have to be defined ultimately in a natural language" (Harris, 1991, 274), as the surging demand for interpretability and explainability in NLP has made evident (see e.g. Danilevsky et al., 2020). Language is consequently not a "code" of "forms" that correlate with some meanings outside of it. It has no "one-to-one" conformity "with some independently discoverable structure of meaning" (Harris, 1954, 152). Instead, it is a system related to, but also independent from, thought (Nevin, 1993, 361–363). While all human activity is meaningful, the particular meanings of language are *constituted* by its form, not *correlated* to it (Nevin, 1993, 394). Meaning thus understood is about departures

from equiprobability in the distribution of these constraints (Harris, 1991, 23). Linguistic inquiry is, in the final instance, about evaluating the efficiency of different grammars and their ability to model the statistical constraints imposed upon the distribution of different linguistic elements. Harris held that language was a "detached pattern" (Harris, 1941, 295)—information that was socially transmissible and hence constitutive of new types of socially shared and conventional meaning (360) (Nevin, 1993, 360, 365)—and linguistics could at best discover different incomplete grammars. Though linguistics might provide insights about meaning and discourse, or about cultural practices, such findings would not bear directly on linguistics *per se*. Indeed, while Harris acknowledges that our sense of word meaning is aided by "extra-linguistic situational information," words "beyond the immediate situation" are "on their own" (Harris, 1991, 368). However, Harris's method is not completely detached: Searching for a method to segment speech, he notes that the similarity of elements "reduces ultimately to the similarity of sound segments under repetition," implemented through "the pair test" in which native speakers are asked to discriminate between sound segments (Harris, 1954, 158–159), producing an observational primitive that is "more easy to control than data on meaning" (e.g. Harris, 1951, 20).

3.2 Variation, or lack thereof

Harris's view of language and linguistics, isolated from the vagaries of social interaction and variation, is obviously difficult to reconcile with a sociolinguistic perspective. As Harris writes in his *Methods of Structural Linguistics* (Harris, 1951, 9), his approach is meant to describe a homogeneous dialect, which "[i]n most cases...presents no problem, since the whole speech of the person or community shows dialectal consistency." Referencing this passage, dialectology pioneer William Labov (2006, 5) argues that "the inconsistency found in most New York City idiolects is so great that the first alternative of Harris is impossible, and the second implausible."[1] In other words, even at the level of the speaker, Harris's idealized, unvarying idiolect does not hold up to empirical scrutiny. Rigorous consideration of factors that Harris would deem extra-linguistic (class, race, interactional roles, etc.) are indeed essential to produce a systematic description of linguistic structure. From the sociolinguistic perspective, Harris's vision of a linguistic science fully isolated from the "intrusion" of non-verbal social life would never obtain the systemicity to which it aspired.

The critique from sociolinguistics highlights more fundamental issues in Harris's structuralist perspective. Language is viewed primarily through the distributional restrictions imposed by convention, rather than "stylistic practice" and the ways in which speakers "make social-semiotic moves, reinterpreting variables and combining and recombining them in a continual process" (Eckert, 2012, 94). We can study changes in discourse, as Harris himself did in an impressive volume on structures in immunological theory over time (Harris et al., 1988), but not how people make those changes, or indeed the way in which "variation itself is systemic" (Deleuze and Guattari, 1987, 93).

3.3 Sublanguages

Harris introduced the concept of sublanguages in his book *Mathematical Structures of Language*, defined as "certain proper subsets of a language [which] may be closed under some or all of the operations defined in the language, and [which] thus constitute a sublanguage of it" (Harris, 1968, 152). The sublanguage is a set of sentences which are a subset of the sentences of the "whole" language. However, the grammatical constraints of the sublanguage are not necessarily those of the whole language; rather, their grammars intersect (Harris 1968, Ch 11; Kittredge and Lehrberger 1982, 1). In application the term has come to refer primarily to the grammar and vocabulary unique to or characteristic of a particular professional or scientific field (e.g. Harris, 1988), an influential concept for early information retrieval research (Sager, 1975, 1981). Harris believed that sublanguages could be neatly identified using the distributional methods of his general linguistic program.

4 Firth's contextual semantics

Something similar to the sociolinguistic critique towards Harris could be articulated from another perspective as well, namely through the work of J.R. Firth. Firth (1890–1960)—professor of General Linguistics at the University College of London and the first holder of a chair in that subject at any British university—independently formulated

[1] New York City is not unique in this regard; it was merely the location of his early pathbreaking work.

a distributional theory of lexical semantics. However, unlike Harris, Firth refused to treat semantics separate from pragmatics, and words apart from their broader "context of situation." (Robins, 1997, 205–208)

Firth never published a fully articulated exposition of his general theory of language (Robins 1997, 216; Thomas 2011, 180) and today, most of his work is out of print and mostly unavailable online. Not understood by "the contemporary scientism" of American descriptivist linguistics and its pioneers like Harris, Firth was mostly ignored on the other side of the Atlantic (Pandit 1970, 280; Palmer 1968, 2). Unlike many of his American contemporaries, Firth did not draw mainly from cognitive psychology and logic—the latter which Firth thought had "taken the heart out of language" (Firth, 1957a, 186)—but from the work of Polish anthropologist Bronislaw Malinowski (Robins, 1997, 211). Here, we focus on the evolution of his thoughts on meaning and collocation, as well as his notions of *context of situation* and *restricted language*.

4.1 Meaning by collocation

To Firth, the purpose of linguistics is to "study meaning in its own terms" (Firth 1968a, 145 ; Senis 2015, 289). The famous phrase about the company that words keep concerned a particular "mode of meaning," "meaning by collocation". Anticipating vocabulary now ubiquitous in NLP, Firth thought that this level of meaning could be found by examining the "habitual collocations" of words and the "word material" in which they are "most characteristically embedded" (Firth, 1957b, 11–12). Meaning by collocation was an abstraction of *syntagmatic relations* (Firth 1957b, 12; Oyelaran 1967, 444) that went beyond "mere juxtaposition," stating instead "an order of mutual expectancy" (Firth, 1957b, 12) and "mutual prehension." While mutual expectancy could be understood similarly to Joos's (1950) conditional probabilities of occurrence or the concept of PMI (Fano, 1949), the notion of "prehension" originates in the work of philosopher and mathematician Alfred North Whitehead (1957, 1968, see also Butt 2013) and concerns the manner in which one entity grasps another and makes it part of its own experience (Christian 1959, 12; Bryant 2011, 136).

Drawing on Whitehead's "modes of thought" (Whitehead 1968; Butt 2001, 1812; Butt 2019, 28), Firth advocated a type of "polysystemic" linguistic analysis that was interested in different, congruent modes of meaning, whether phonetic, phonological, syntactic, or semantic, but always situated in broader social context (Firth 1957b, 27, 30 ; Robins 1997, 214). In stark contrast to Harris, Firth explicitly rejected any efforts to create "unity in linguistics" or one system of analysis. Citing the later Wittgenstein (1953; Firth 1957b, 11), Firth was mainly interested in the concrete use of language, reversing the Saussurian schema (De Saussure, 2011) in which language (*la langue*) is a system "external to and on a different plane from individual phenonema," including the concrete instances (*la parole*) of language use (Firth 1950, 44–45; Firth 1949, 144). While his final ideas matured considerably later, Firth initially articulated his ideas about semantics in two 1935 papers, one on semantics (Firth, 1935a) and one on phonology (Firth, 1935b), using the term "contextual distribution" in both (see in particular Firth 1935a, 55). However, Firth (1957b, 18) ultimately disavowed this distributional theory—which was not too dissimilar from Harris's—as "useful" but inadequate to act as the "main principle" in a theory "of structures involving the statement of the values of the elements of structure by reference to systems."

By distinguishing between system (syntagm) and structure (paradigm), Firth wanted to highlight two operational principles necessary for meaning by collocation: (1) substitution within "the same level of abstraction," and (2) commutation across different levels (Robins, 1953, 140). Only substitution that does not produce commutation in a sequence, indicates similarity of value or function (Firth 1968b, 23; Firth 1957b, 15). I.e. two words are only substitutable—and hence similar in function and meaning—if we ensure that their values do not commute across a particular sentence. Substitution, then, does not equal synonymy. Take, for example, the following two phrases containing (a) prepositional and (b) adverbial uses of the word "by" :

(a) They go by night.
(b) They go by night after night.

Now, "by" could be replaced by the word "past" without commuting the meaning of the other words in (b). However, replacing "by" with "past" would commute with the rest of (a) in an impossible way (Firth, 1968b, 23–24). This demonstrates how substitution concerns the relationship between "by"

and "past" as two elements at the same level of analysis—i.e., lexical units—but in order to account for commutation, we need to look beyond this level to other levels of abstraction.[2]

4.2 Context and connection

Diverging from structuralist linguistics, Firth suggested that a text should always be given a "renewal of connection with experience" (Firth, 1957b, 29). This notion of meaning was influenced by Malinowski, for whom Firth worked as an assistant early in his career (Plug, 2008, 346) and from whom he borrowed the notion of "context of situation" (Firth 1935a, Robins 1997, 211). In Malinwoski's view, meaning was more than just a dyadic relationship between a word and its referent, "a multidimensional and functional set of relations between the word in its sentence and the context of its occurrence" (Robins, 1971, 35). However, while Malinowski's view on meaning was entirely functional and hyper-local, Firth employed the notion of "context of situation" as a necessary abstraction, not as a shorthand for things in themselves (Firth 1950, 7 ; Palmer 1968, 6). Context of situation is derived from an analytical choice, "a set of categories in ordered relations abstracted from the life of man in the flux of events, from personality in society," prehending something of importance and bracketing the rest. It is, then, not necessarily about restricting the meaning of every utterance to a specific time and place, but about defining "an abstract set of semantically relevant categories, abstracted from multitudes of actual situations, to which unique particulars could be referred." (Robins, 1971, 41–42) Firth called for the linguist to focus on "attested language text duly recorded" (Firth, 1957b, 29–30), accounting for both the interior relations of the text as well as its associated context of situation *and* its interior relations. (Firth, 1957b, 30)

Firth was famously opaque with the exact operationalization of his concepts, including context of situation, but he did provide a detailed list of the different contextual elements that a linguist should bring into relation during analysis (Firth, 1950, 43):

A. The relevant features of participants: persons, personalities.
   (i) The verbal action of the participants.
   (ii) The non-verbal action of the participants.
B. The relevant objects.
C. The effect of the verbal action.

In conclusion: Firth's famous quote itself refers to collocation, while his notion of "context" implies something much broader, "the whole conceptual meaning" (Firth, 1957b, 11). Context is the ground against which the figure of the text must be understood, no matter (per Harris) how "detached" its pattern might be (e.g. Auer, 1996). Without context, collocation captures only one narrow "mode of meaning."

4.3 Restricted languages

Expanding upon his functional understanding of language, Firth developed his notion of "restricted languages" in the 1950s (1957b, 10; Léon 2007, 7). In a posthumously published essay, he (1968c, 207) describes social actors as "collect[ing] a varied repertory of interlocking roles" corresponding to a "constellation of restricted languages." As people shift between locally contextualized roles, they draw upon their "repertory" of restricted languages with specialized vocabulary and discursive styles that both reflect and constitute these contexts. Thus one might speak of a "restricted language of science, sport, defense, industry, aviation, military services, commerce, law and civil administration, politics, literature, etc." (Léon, 2008, 261). As such, the concept of restricted language is now generally seen as a precursor to the concept of "register," which was taken up by subsequent sociolinguists and linguistic anthropologists (e.g. Halliday, 1968; Gumperz and Hymes, 1972; Agha, 2005).

In proposing restricted languages as the proper object of descriptive linguistic analysis, Firth was making a broader theoretical point against, on the one hand, "the monosystemic view of language" of neo-Bloomfieldians like Harris and "pointless discussions on metalanguage" on the other, for metalanguage could be reanalyzed as a "restricted language of linguistics" itself (Léon, 2007, 9). Simply put, a descriptive linguistics which privileges restricted languages also necessarily privileges contexts of situation as an essential dimension of variation that allows social meaning to inhere in language.

5 Discussion: Words in new company

Often cited, together or separately, with the same purpose of justifying a distributional view on se-

[2] A useful analogy might be the way in which BERT handles different aspects of language at different layers of the model (Tenney et al., 2019, e.g.). However, no matter how large, a language model like BERT does not account for the context of situation.

mantics, Firth and Harris nonetheless present vastly different views on language and meaning. In Harris we find a rigorous formalism that treats language as a "detached pattern." Language is not a "code", but functions instead as a particular system of meaning. Firth, by comparison, left a much more scattered legacy and many of his insights were only systematized and formalized by his students. Firth and Harris shared a concern about the dependence between the languages *of* and *under* description, but drew different conclusions regarding the tension between form and meaning in language. If Harris responded to this conundrum by creating one system without intrusion from extra-linguistic factors, Firth called for an investigation of language as a "spectrum" with different modes of meaning that had to be addressed through multiple levels of analysis, without a "unity of linguistics." Analysis should start by stating the context of situation and proceed from there to decide which other levels of analysis are relevant. Firth's distributional theory has been contrasted unfavourably as "frequency based" to the "recursive dependencies" developed by Harris (Habert and Zweigenbaum, 2002, 205). For Harris, the meaning of a word depends on its set (Harris, 1991, 17) such that, for example, the words "divide" and "multiply" operate on the word "cell" (and vice versa) in the same way, producing essentially the same meaning (Harris 1988, 62; see also Harris 1991, 5–6). However, Firth's final method of substitution and commutation establishes complex, multidimensional criteria for distributional contrast as well as a framework for understanding polysemy. Though less formalized than Harris, Firth's approach ultimatelly also treats linguistic elements as operators with specific sets of elements in a complex hierarchy, *including* the extra-linguistic context of situation.

Nonetheless, from the perspective of empirical work, especially decades after their time, Firth and Harris also share similarities. Both rejected the mentalism that was so prevalent during their time. They were both revived as empiricist originators during the rise of statistical learning in the 1990s and their respective work on restricted languages and sublanguages was important for the rise of corpus linguistics and largely conflated in service of practical concerns (Léon, 2008). They both included in their theories what we might call a "reality principle," a final arbiter of meaning outside of form: the pair test for Harris, the context of situation for Firth. However, as NLP tries to move beyond the written corpus, we might reassess the details and differences between Firth and Harris in order to understand the practical limits and possibilities of the distributional view on language.

In light of recent calls to extend the "world scope" (Bisk et al., 2020) of NLP and to move towards pragmatic notions of meaning, we might find Harris's rigorous but rigid system somewhat limiting and Firth's appeals to different contexts helpful. Though Firth warned us against overextending linguistics, he was generous with the company that words could keep. They mingled with each other, but also with events, objects, people, and indexical features such as time and space. And if NLP is ready to move beyond the corpus, then even Harris might acknowledge that when modeling language in "the immediate situation"—whether in online interactions or face-to-face communication—words are *not* on their own, that to judge the meaning of a combination of words, we can summon "the aid of some of the extra-linguistic situational information" (Harris, 1991, 368). In the next two subsections, we consider the ways in which NLP is already doing this, in order to highlight some existing strategies for broader contextualization.

5.1 Comparative stratification

Corpus linguistics emerges from the question of what kind of company words keep, depending on their context. The issue of context was motivated the introduction of corpus linguistics both for students of Firth—who considered restricted languages as a way of handling context—and for Harris's sublanguages, "concrete discourse structures, which could be applied to real scientific texts, contextually situated and suitable for being processed automatically" (Léon, 2021, 149–150). However, beyond just studying the restricted corpus, we might also consider the ways in which large datasets can be "stratified"—a type of implicit factorization, if you will—systematically dividing them into sub-corpora that are rigorously and statistically studied in relation to each other. Here, the company that words keep among each other is *limited* for analytical purposes, but in a manner that implicitly tell us what kind of context different kinds of words keep as company.

Diachronic embeddings are especially representative of this approach. By stratifying timestamped data into a number of intervals, training a separate

model for each interval and then aligning the embeddings using either "second-order embeddings" or methods such as linear transformations (Kutuzov et al., 2018), the analyst can effectively give words a temporal "context of situation." This general approach works with both static (Hamilton et al., 2016) and contextual embeddings (Martinc et al., 2020). While most commonly used in the literature on semantic shifts (e.g., Garg et al., 2018; Kozlowski et al., 2019; Mendelsohn et al., 2020), it could plausibly be used to stratify a dataset using other variables such as space (e.g. Yan et al., 2017), online communities (Lucy and Bamman, 2021), persons (Yao et al., 2020), or domains (Spinde et al., 2021). Words still only keep the company of other words, but by limiting their company we can implicitly introduce other participants in the analysis.[3]

5.2 Syntagmatic extension

Recall that for Firth, meaning by collocation and considerations of a "typical context of situation" were exercises in abstraction, with collocation being an abstraction at the syntagmatic level. Instead of restricting the company words keep, we might follow Firth's recommendation to consider them in wider company "of the same abstract nature" (Firth, 1950, 7). In vector semantics, this would imply that we explicitly introduce different contextual factors in the *same* vector space with our words, giving all participants in the company ontological equality.

The paragraph vectors introduced by Le and Mikolov (2014) as an extension of their skipgram algorithm (Mikolov et al., 2013) are representative for this approach. In practice, this method extends the syntagmatic chain of words by introducing a vector for the document as a new paradigmatic element. This type of "global context" (Grbovic and Cheng, 2018) could in principle be anything and include several paradigmatic elements, as we can see in the research on multi-modal embeddings (e.g. Baroni, 2016).[4] For example, models have been developed that include demographic (Garimella et al., 2017) or persona (Li et al., 2016) vectors in the embedding space, such that intra-textual relations are accompanied by information about speakers' social categories. However, implemented with static embeddings and without some additional grammar restrictions, these context vectors essentially add only a "bag of contexts." For static embeddings, additional grammar constraints could be introduced, as had been done in research on Point-of-Interest (POI) data in the the domain of geosemantics, where researchers constrained contextual vectors using spatial variograms (Yan et al., 2017). Beyond static embeddings, large-language models and their dynamic embeddings could either be pretrained (with the appropriate dataset) or at least finetuned on data with text associated to different contextual variables. This would realize the proposal that Halliday made in the early 1990s when he suggested an extension of the language modeling schema from the early work of Shannon, to a model with "global probabilities, those of the grammar of English, and the locally conditioned probabilities, those of this or that particular register" (Halliday, 1991, 37).

6 Conclusions

This paper revisited the theories of the two most well-known progenitors of distributional semantics. Recognizing the ongoing debate on how to bring NLP in general and language modeling in particular beyond the corpus, we provide a thorough account of the distributional theories that are most frequently evoked in order to justify the modeling of meaning through departures from equiprobability in the company that words keep. Comparing the work of Harris and Firth—who both published their major work before the rise of the internet and its corpora—we find two distinct theories of distribution: one formal and mathematical, treating language as a particular type of detached information, another more schematic and anthropological, treating language as a functional spectrum which always emanates from a particular context of situation. The legacies of both Firth and Harris can be seen in the current paradigm of corpus linguistics, but in the domain of distributional semantics, it is Harris's ethos that dominates, despite Firth providing its most famous tagline. Moving forward, we suggest that semantic modeling take more cues from Firth, and consider the context of situation and the wide variety of company that words can keep as crucial challenges with which the field must grapple.

[3] In a very broad sense, the trend of pretraining large language models and then finetuning them on specific datasets is of course also an admission of the importance of "context of situation."

[4] Firth himself (1957b, 26) recommended accompanying word definitions and collocational information with pictures.

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